[NAME OF THE UNIVERSITY / INSTITUTE]

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A PROJECT REPORT ON

RED SEA SHIPPING DISRUPTION: DIAGNOSING THE CRISIS AND DESIGNING A RESILIENT SUPPLY CHAIN RESPONSE

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LIST OF ABBREVIATIONS

3PL	Third-party logistics provider
AIS	Automatic Identification System (ship tracking)
ARIMA	Autoregressive integrated moving average
CIF	Cost, insurance and freight
CO2	Carbon dioxide
EBITDA	Earnings before interest, tax, depreciation and amortisation
EU	European Union
EU ETS	EU Emissions Trading System
FEU	Forty-foot equivalent unit (container)
FOB	Free on board
FY	Financial year (India: April–March)
HAC	Heteroskedasticity and autocorrelation consistent
HS	Harmonized System (product codes)
IMEC	India–Middle East–Europe Economic Corridor
IMF	International Monetary Fund
INSTC	International North–South Transport Corridor
JWC	Joint War Committee (Lloyd's market)
KPI	Key performance indicator
LP	Linear programming
OLS	Ordinary least squares
OTIF	On time in full
S&OP	Sales and operations planning
SS	Safety stock
TLC	Total landed cost
UNCTAD	United Nations Conference on Trade and Development
WCI	World Container Index (Drewry)

EXECUTIVE SUMMARY

Interactive dashboard: Red_Sea_Strategy_Dashboard.html (provided with this report; keep it in the same folder as this report and open it in any web browser). It shows the real data, strategy comparison, what-if controls and advanced analysis behind this report.

Using real data from IMF PortWatch, Drewry and India's Ministry of Commerce, plus four annual reports of Gokaldas Exports, this project shows that the Red Sea crisis is still raising Asia–Europe supply chain costs three years on. An Adaptive Resilience Strategy (contracted Cape capacity, an EU buffer warehouse, rule-based expediting and a pre-agreed trigger to return to Suez) cuts a model buyer's annual landed logistics cost from $7.52M to $5.18M (31%).

Since November 2023, Houthi attacks near the Bab-el-Mandeb Strait have pushed container shipping around the Cape of Good Hope. PortWatch data show container transits through Bab el-Mandeb down 72% and through the Cape up 234% (January–September 2026 against January–October 2023). Drewry's Shanghai–Rotterdam rate averaged $4,941 per FEU in 2024 (3.4 times pre-crisis), peaked at $8,267 in July 2024 and rebounded to $4,933 in mid-2026.

The project diagnoses the problem with root-cause tools (fishbone, 5 Whys, PESTLE, SWOT), documents firm-level effects at Gokaldas Exports, and evaluates eight strategies for a model European buyer of Indian apparel (1,200 FEU a year) on total landed cost, lead time, reliability, risk, sustainability and feasibility.

Key findings

- India's apparel exports to the EU did not collapse: FY26 was 1.7% above FY23. The crisis showed up as cost, delay and inventory rather than lost volume.
- At Gokaldas Exports, freight cost more than doubled in FY25 (₹18.4 crore to ₹40.1 crore) but its management attributes this to production disruptions, not the Red Sea, because buyers on FOB terms paid the extra ocean freight; a second exporter, KPR Mill, shows no freight spike at all.
- The status quo (spot bookings via the Cape) costs the model buyer $6,270 per FEU at the latest 52-week average rate of $3,012, and late-delivery losses ($2.45M a year) are its biggest hidden cost.
- The Adaptive Resilience Strategy scores highest (3.90 out of 5), saves $1.4M–$3.7M a year at every rate level observed since 2023, and pays back in about 2.5 months. In a 10,000-run Monte Carlo simulation it saves money in 99.99% of runs (90% range $1.16M–$3.86M).

- Suez container transits in October 2026 reached their highest level since December 2023, so the trigger to return may be close; the strategy is built to capture that saving safely.

Note on data: all market data are real (Section 4.2 and References). The model buyer's operating parameters, such as volume, product values and late-delivery loss rates, are stated assumptions; Sections 7.4 and 10.1 show how they affect the results.

CHAPTER 1
INTRODUCTION

The Red Sea crisis is the largest sustained disruption to a global shipping chokepoint since the 2021 Ever Given grounding, and unlike that six-day blockage it has lasted for years. It shows how a regional security event becomes a global cost, inventory and emissions problem for firms with no direct link to the conflict.

1.1 Problem statement

Since late 2023, the Red Sea–Suez Canal corridor, which normally carries about 12–15% of world seaborne trade and roughly 30% of global container traffic, has been effectively closed to most container shipping because of armed attacks on vessels. Diverting around the Cape of Good Hope has increased transit times, freight costs, lead-time variability and emissions. Firms on Asia/India–Europe lanes face higher landed costs, stockouts, missed seasonal windows and working-capital strain.

The core operations problem: how should a firm dependent on this corridor redesign its logistics and inventory policy to minimise total landed cost and service risk while the corridor's future remains uncertain?

1.2 Research objectives

1. Describe the disruption, its timeline and its current status.
2. Quantify its impact on cost, lead time, reliability, inventory and emissions.
3. Identify root causes and the structural vulnerabilities that turned a regional event into a global problem.
4. Develop and evaluate alternative response strategies using a total-landed-cost model and multi-criteria scoring.
5. Recommend a strategy with an implementation roadmap, KPIs and a risk register.

1.3 Research questions

- RQ1: What are the measurable operational effects of the Red Sea disruption on Asia–Europe supply chains?
- RQ2: Why did the disruption propagate so widely (root causes and amplifiers)?
- RQ3: Which routing and inventory strategy minimises total landed cost and risk for a representative importer?
- RQ4: How should that strategy adapt if the corridor reopens or the crisis worsens?

1.4 Scope

Table 1.1: Scope of the study

Dimension	In scope	Out of scope
Geography	India and Asia to North Europe container lanes	Tanker and bulk markets (mentioned only for context)
Industry	Apparel and consumer goods (time-sensitive, seasonal)	Bulk commodities, energy
Period	November 2023 to mid-2026, with scenarios to 2027	Pre-2021 history beyond context
Perspective	Importing firm (shipper); other stakeholders analysed	Military and diplomatic policy recommendations

1.5 Significance

The crisis is a live test of supply-chain resilience theory. Its lessons apply to other chokepoints (Strait of Hormuz, Strait of Malacca, Panama Canal, Taiwan Strait), making the framework in this report reusable beyond the Red Sea.

CHAPTER 2
BACKGROUND AND CURRENT SITUATION

The Suez Canal and the Bab-el-Mandeb Strait form the shortest sea link between Asia and Europe, and no land or sea alternative of similar capacity exists. The canal handled roughly 20,000–26,000 transits a year before the crisis and earned Egypt about $10 billion in fiscal 2022/23.

2.1 Why the corridor matters

- **Distance:** Shanghai to Rotterdam is about 10,500 nm via Suez and about 13,800 nm via the Cape; India's west coast to Rotterdam rises from about 6,300 nm to roughly 10,500 nm.
- **Trade share:** about 12–15% of world trade by volume and around 30% of global container traffic used the route.
- **India's exposure:** a large majority of India's merchandise trade with Europe, North Africa and the US East Coast moved through the Red Sea, so Indian exporters of apparel, engineering goods, chemicals and agricultural products were among the most exposed.

2.2 Timeline of the crisis

Table 2.1: Timeline of the Red Sea crisis, November 2023 to October 2026

Date	Event	Supply chain effect
Oct 2026	Suez container transits reach 13.6 a day over 1–11 October, the highest since December 2023; Cape transits fall to ~15.6 (PortWatch)	First data signal of a partial return
Jun–Aug 2026	Shanghai–Rotterdam rate rebounds from ~$2,550 to a 2026 high of $4,933 (Drewry)	Second cost spike; on its 25 May 2026 investor call Gokaldas linked higher shipping and input costs to the U.S.–Iran war
Oct 2025	Gaza ceasefire; Houthis signal a pause; spot rate falls to a post-crisis low of $1,577 (Drewry, 9 Oct)	Overcapacity pushes rates down; carriers stay on the Cape
Jul 2025	Bulk carriers Magic Seas and Eternity C sunk	Return postponed
May 2025	US–Houthi ceasefire brokered by Oman	Little change for container lines
Jan 2025	First Gaza ceasefire	Brief optimism; Cape routing continues
18 Jul 2024	Spot rate peaks at $8,267 per FEU (Drewry)	Peak cost pressure across the industry
Mar–Jun 2024	Rubymar and Tutor sunk; first crew deaths (True Confidence)	Route seen as unsafe for the medium term

Date	Event	Supply chain effect
Jan 2024	US and UK strikes begin; Bab el-Mandeb container transits fall to 4.0 a day (PortWatch)	Diversion becomes near-universal
Dec 2023	Maersk, MSC, Hapag-Lloyd and CMA CGM pause Red Sea transits; Operation Prosperity Guardian	Mass rerouting via the Cape begins
19 Nov 2023	Houthis seize the car carrier Galaxy Leader	Start of the crisis

2.3 Current situation

As of October 2026, the disruption is chronic but may be turning. The Cape of Good Hope still carries over three times its pre-crisis container traffic (19.7 ships a day in January–September 2026), yet Suez container transits rose sharply in September–October 2026. Freight rates remain 2.4 times pre-crisis ($3,522 on 15 October 2026). Timeline events without a named data source come from news coverage; confirm their dates against Reuters or Lloyd's List before submission.

CHAPTER 3
LITERATURE REVIEW

The literature agrees that resilience comes from a mix of redundancy (buffers) and flexibility (options to switch), and that the right mix depends on product type and disruption duration. Five strands frame this project.

Table 3.1: Summary of the literature reviewed

Theme	Key authors	Core idea	Relevance to Red Sea crisis
Supply chain resilience	Sheffi & Rice (2005); Christopher & Peck (2004); Ponomarov & Holcomb (2009)	Resilience = ability to prepare, respond and recover; built through redundancy and flexibility	Frames the choice between buffer stock (redundancy) and multi-route options (flexibility)
Risk management	Tang (2006); Chopra & Sodhi (2004)	Match mitigation to risk type; avoid one-size-fits-all buffers	Chokepoint risk is low-frequency, high-impact and persistent, needing structural responses
Matching supply chain to product	Fisher (1997)	Functional products need efficient chains; innovative products need responsive chains	Supports segmenting fashion (responsive) vs basics (efficient) in routing decisions
Bullwhip effect	Lee, Padmanabhan & Whang (1997)	Order variability amplifies upstream	Panic ordering and early peak season in 2024 inflated rates and congestion
Inventory theory	Silver, Pyke & Thomas (2017); Chopra & Meindl	Safety stock grows with lead time and lead-time variability	Basis for this project's safety-stock and pipeline-inventory model
Ripple effect and disruption propagation	Ivanov (2017); Dolgui, Ivanov & Sokolov (2018)	Disruptions spread across networks and persist after the trigger ends	Explains equipment shortages and port congestion far from the Red Sea
Real options in operations	Trigeorgis (1996); Kouvelis et al.	Flexibility has measurable value under uncertainty	Justifies paying for a pre-agreed Suez return trigger and flexible contracts

Research gap. Most studies analyse disruptions that are short (Ever Given) or global (COVID-19). Fewer address a *persistent, partial, reversible* chokepoint closure where the firm must decide not only how to cope but when to switch back. This project contributes a decision framework combining total landed cost, conditional expediting and trigger-based route switching for this scenario.

Citations are from the author's knowledge base; verify page and edition details in Google Scholar before submission.

CHAPTER 4
RESEARCH METHODOLOGY

The study uses a mixed-method design: secondary data establishes the industry-level impact, and a quantitative case model evaluates strategies for one representative firm.

4.1 Research design

- **Type:** descriptive (impact) + analytical (root cause) + prescriptive (solution modelling).
- **Approach:** single illustrative case study with scenario and sensitivity analysis.

4.2 Data sources (secondary)

Table 4.1: Secondary data sources

Dataset	Coverage	Used for
IMF PortWatch, daily chokepoint transit calls (Suez Canal, Bab el-Mandeb Strait, Cape of Good Hope)	1 Jan 2023 – 11 Oct 2026, daily, by vessel type	Traffic diversion (5.1) and the return trigger (9.2)
Drewry World Container Index, Shanghai–Rotterdam, compiled from weekly releases reprinted by Hellenic Shipping News, JOC, AJOT, FreshPlaza and MTS Insights	195 weeks, 5 Jan 2023 – 15 Oct 2026	Freight trend (5.2) and cost-model inputs (7)
Ministry of Commerce, TradeStat: India's monthly exports by country, HS 61, 62, 84, 85	April 2022 – July 2026, 237 countries	Trade impact (5.3)
Gokaldas Exports Ltd annual reports, FY2022-23 to FY2025-26	Audited standalone financials, FY22–FY26	Firm-level evidence (5.4)
UNCTAD Red Sea brief (2024) and Review of Maritime Transport; Suez Canal Authority; Sea-Intelligence	Context figures	Background, emissions, reliability

Additional company sources: Gokaldas Exports quarterly investor presentations, financial results and earnings-call transcripts (Q1 FY24 to Q4 FY26, 36 documents), and KPR Mill annual reports FY2022-23 to FY2025-26.

4.3 Case setting

The analysis uses two cases. **Gokaldas Exports Ltd** is the real exporter case (Section 5.4): its audited figures show how the crisis reached an Indian supplier. **EuroThread** is a model European buyer of Indian apparel, used for the cost model in Sections 7–8, because routing, contract and buffer decisions sit with the party paying ocean freight. Its freight inputs come from the real Drewry series; its operating parameters are stated assumptions.

Table 4.2: Case-setting parameters and their basis

Parameter	Value	Basis
Pre-crisis spot rate (via Suez)	$1,474 per FEU	**Real:** Drewry average, 5 Jan – 9 Nov 2023
Base-case Cape spot rate	$3,012 per FEU	**Real:** Drewry average, 52 weeks to 15 Oct 2026
Crisis-year spot rate	$4,941 per FEU	**Real:** Drewry 2024 average
Annual volume	1,200 FEU	Assumption: mid-size European retailer
A-class (fashion) share	15% of volume, $80,000 per FEU	Assumption (ABC analysis)
B/C-class (basics) share	85% of volume, ~$44,700 per FEU	Assumption; total flow $60M a year
Contract rate	78.6% of spot	Assumption: typical contract discount
Return-to-Suez rate	1.6 × pre-crisis + 0.5% war-risk premium	Assumption
Transit time, Suez / Cape	22 / 34 days	Carrier schedules, India–North Europe
Lead-time standard deviation, Suez / Cape	2 / 6 days	Assumption, consistent with reliability data
Holding cost rate	20% per year	Assumption, apparel norm
Target service level	95% (z = 1.65)	Assumption
Late-delivery loss	A 25%, B/C 10% of value; chance of lateness 30% on Cape spot, 8% with buffer	Assumption; tested in 7.4
Daily demand standard deviation	0.6 FEU	Assumption
EU buffer warehouse (3PL)	$180,000 per year	Assumption, quoted range

4.4 Analytical tools

- Root cause: Ishikawa (fishbone), 5 Whys, PESTLE, SWOT.
- Quantitative: total landed cost (TLC) model, safety-stock model, expected late-delivery loss, break-even expediting rule.
- Decision: weighted scoring model (multi-criteria), scenario analysis (three futures), sensitivity analysis.
- Visualisation: interactive dashboard (linked separately).

4.5 Key formulas

Total landed logistics cost per year:

$$\mathbf{TLC = \Sigma\,[\; n \times (F + I \times V) + n \times V \times h \times (LT \div 365) + SS \times V \times h + n \times P(late) \times \pi \times V\;] + Fixed\ costs}$$

(the sum Σ runs over the product classes A and B/C)

Safety stock with demand and lead-time uncertainty:

$$\mathbf{SS = z \times \sqrt{(\; LT \times \sigma d^2 + \bar{d}^2 \times \sigma LT^2\;)}}$$

(z = service-level factor, σd = standard deviation of daily demand, $\bar{d}$ = average daily demand, σLT = standard deviation of lead time)

Expedite a shipment by sea-air only if the avoided loss exceeds the freight premium:

$$\mathbf{Expedite\ only\ if:\ V \times \pi \times \Delta P(late) > F(sea\text{-}air) - F(sea)}$$

(ΔP(late) = the fall in the chance of a late delivery that expediting achieves)

Here n is shipments, F freight per FEU, I insurance rate, V cargo value, h holding rate, LT lead time, P(late) probability of missing the sales window, and π the loss rate when late.

4.6 Advanced analytical methods

Five further methods test and extend the core model; results are in Sections 5.8 and 7.5–7.7.

Table 4.3: Advanced analytical methods

Method	Question it answers	Data
Welch t-test and Mann–Whitney U test	Did transits and freight rates change significantly after the crisis began?	PortWatch daily, Drewry weekly
OLS regression with HAC (Newey–West) standard errors	Did higher freight rates reduce India's apparel exports to the EU?	TradeStat monthly, Drewry monthly
Monte Carlo simulation (10,000 runs)	How certain is the saving when inputs are uncertain?	Drewry weekly (bootstrap) + assumed distributions
Linear programming (HiGHS solver)	What is the cheapest mix of transport modes for each product class?	Model parameters, real rates
ARIMA time-series forecast	Where are rates heading, and can they be predicted?	Drewry weekly

Regression model:

$$\mathbf{\ln(EU_t) = \alpha + \beta \times \ln(Rate_{t-1}) + \gamma \times t + \Sigma\, \delta_m \times Month_m + \varepsilon_t}$$

(EU_t = apparel exports to the EU in month t, $Rate_{t-1}$ = previous month's freight rate, t = time trend, $Month_m$ = month-of-year dummies, β = freight elasticity)

Linear programme (x = FEU of product class s sent by mode k; c = freight + pipeline holding + expected late-delivery loss per FEU; p = probability of a late delivery):

$$\mathbf{Minimise:\ Total\ cost = \Sigma\, c_{sk} \times x_{sk}}$$

Subject to:

1. $\Sigma_k\, x_{sk} = n_s$ for each product class s (all volume is shipped)
2. $\Sigma\, p_k \times x_{sk} \leq (1 - \text{OTIF}) \times N$ (on-time delivery target)
3. $\Sigma\, x(\text{sea-air}) \leq 0.1 \times N$ (sea-air capacity limit)
4. $\Sigma\, x(\text{contract}) \leq 0.8 \times N$ (contract capacity limit)
5. $x_{sk} \geq 0$ (no negative shipments)

(N = total FEU per year = 1,200)

CHAPTER 5
IMPACT ANALYSIS

Real data confirm a large and persistent shock. Container transits through Bab el-Mandeb fell 72% in 2024 against January–October 2023 and were still 72% lower in 2026, while Drewry's Shanghai–Rotterdam rate averaged $4,941 per FEU in 2024, 3.4 times the pre-crisis average of $1,474.

5.1 Ship traffic moved from the Red Sea to the Cape

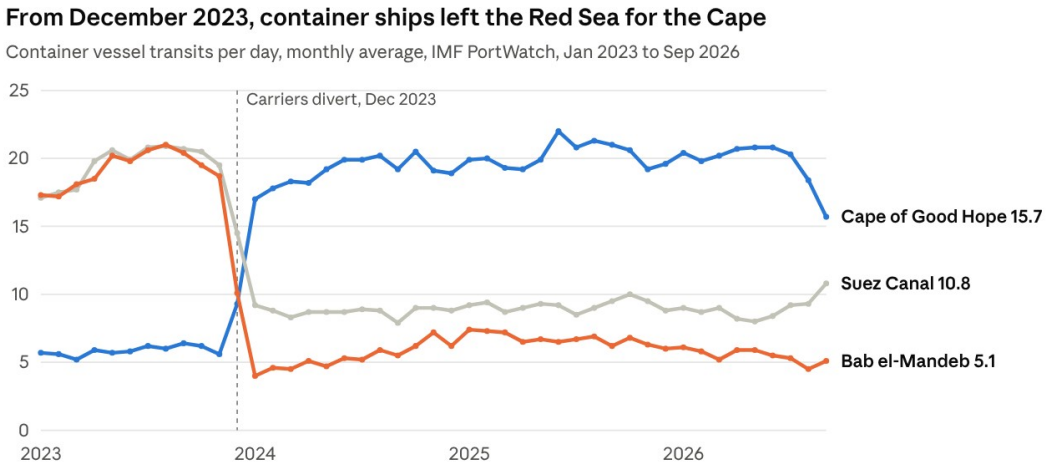


Figure 5.1: Container vessel transits per day by chokepoint, January 2023 to September 2026
Source: IMF PortWatch, daily chokepoint transit calls, averaged by month.

Table 5.1: Container transits per day by chokepoint

Chokepoint (container ships per day)	Jan–Oct 2023	2024	2025	Jan–Sep 2026	2026 vs 2023
Bab el-Mandeb Strait	19.3	5.4	6.7	5.5	−72%
Suez Canal	19.6	8.7	9.2	9.0	−54%
Cape of Good Hope	5.9	19.0	20.2	19.7	+234%

All vessel types show the same pattern: total transits through Bab el-Mandeb fell from 74.6 to 32.5 a day in 2024 (−56%), and the Cape's rose from 48.9 to 87.0 (+78%). Source: IMF PortWatch, daily data, 1 January 2023 to 11 October 2026.

An early sign of return. Suez container transits rose to 10.8 a day in September 2026 and 13.6 a day over 1–11 October 2026, the highest since December 2023, while Cape transits fell to about 15.6. Bab el-Mandeb container traffic has not recovered (about 5–6 a day), so this is a first signal, not a full return. It is exactly the situation the trigger playbook in Section 9.2 is built for.

5.2 Freight rates spiked, eased and rebounded

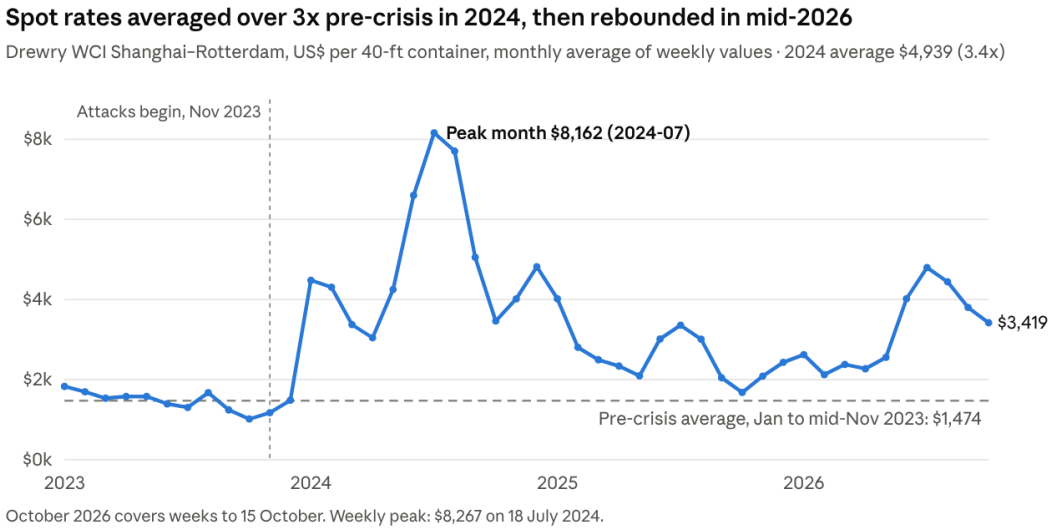


Figure 5.2: Drewry WCI Shanghai–Rotterdam spot rate, monthly average, January 2023 to October 2026

Source: Drewry World Container Index (195 weekly values), compiled from releases reprinted by Hellenic Shipping News, JOC, AJOT, FreshPlaza and MTS Insights.

The weekly series shows three phases. Rates tripled within weeks of the diversions (monthly average $4,480 in January 2024), peaked at $8,267 in the week of 18 July 2024, and fell to a low of $1,577 on 9 October 2025 as new ships absorbed the longer route. A second spike took rates from about $2,550 in May 2026 to a 2026 high of $4,933. Gokaldas Exports' FY26 annual report mentions geopolitical disruptions "including the U.S.-Iran conflict" in 2026, and on its 25 May 2026 investor call management said the U.S.–Iran war had raised input costs and that "shipping costs have increased as well"; confirm details from news coverage. The latest value, $3,522 on 15 October 2026, is 2.4 times the pre-crisis level.

Table 5.2: Drewry WCI Shanghai–Rotterdam spot rate by year

Year	Average ($/FEU)	Low	High	Weeks
2023	1,457	1,004	1,888	51
2024	4,941	2,989	8,267	51
2025	2,627	1,577	4,774	52
2026 (to 15 Oct)	3,253	2,052	4,933	41

Data quality. 181 of the 195 weekly values are reported directly in Drewry releases. The other 14 are carried forward ("rate stable") or back-calculated from a reported weekly change; each is flagged in the dataset and shown as a hollow point in the dashboard.

5.3 India's exports held up; the cost moved into freight and inventory

Table 5.3: India's apparel exports by market, FY23 to FY26

India's apparel exports, HS 61+62 (US$ million)	FY23	FY24	FY25	FY26	FY26 vs FY23
European Union (27)	4,584	4,079	4,554	4,661	+1.7%
United Kingdom	1,470	1,329	1,433	1,500	+2.0%
United States	5,411	4,719	5,333	4,827	−10.8%
Rest of world	4,738	4,418	4,687	4,826	+1.9%
Total	**16,203**	**14,545**	**16,007**	**15,814**	**−2.4%**

Apparel exports to the EU fell 11% in FY24, but the fall began before the crisis: April–November 2023 was 15% below the same months of 2022 because of weak European demand, while the crisis months December 2023–March 2024 were only 4% below a year earlier. By FY26, EU apparel exports were 1.7% above FY23, and engineering exports to the EU (HS 84+85) rose from $12.9bn in FY23 to $16.3bn in FY25.

The implication matters for the solution: trade adapted rather than stopped. The crisis shows up mainly as cost, delay and inventory, not lost volume, so the response must target landed cost and working capital. The US fall in FY26 coincides with US tariffs, a separate shock. Source: Ministry of Commerce, TradeStat, monthly exports by country, April 2022 to July 2026 (Indian fiscal year April–March).

5.4 Firm-level evidence: Gokaldas Exports Ltd

Gokaldas Exports (Bengaluru), one of India's largest listed apparel exporters, provides real company evidence. Figures are standalone (Indian operations), taken from its audited annual reports for FY23–FY26.

Table 5.4: Gokaldas Exports: standalone financial indicators, FY22 to FY26

Year	Revenue (₹ cr)	Clearing, forwarding and freight (₹ cr)	Freight, % of revenue	Inventory (₹ cr)	Inventory days (on revenue)	US share of revenue	Europe share (named markets)
FY22	1,789.1	32.1	1.79%	432.5	88	75.0%	3.0%
FY23	2,221.1	20.4	0.92%	292.3	48	74.4%	3.3%
FY24	2,139.1	18.4	0.86%	362.7	62	70.0%	5.1%
FY25	2,476.7	40.1	1.62%	458.1	68	68.4%	5.1%
FY26	2,693.0	32.0	1.19%	567.9	77	67.0%	7.8%

FY24 covers April 2023–March 2024 (the crisis began in month eight); FY25 covers April 2024–March 2025 and contains the July 2024 rate peak. Europe share sums the European

countries the reports name (Netherlands, UK, France, Belgium, Germany, Denmark), so it is a lower bound.

- **Freight cost more than doubled in FY25** (+118%), from 0.86% to 1.62% of revenue. **But management attributes this to production problems, not the Red Sea.** Staff shortages during India's April–May 2024 elections "resulted in production rescheduling, overtime work, delayed shipments, order cancellation and airfreight to compensate the customers", an EBITDA hit of ₹12.6 crore in Q1 FY25. Worker unrest after acquiring Atraco (Kenya) caused "continued air freights, which happened until August" 2024.
- **Ocean-freight cost was passed to buyers.** On the 2 February 2024 investor call, Managing Director S. Ganapathi said: "The cost of -- increased cost of logistics is being borne by the customers, and it will continue to be borne by the customers", adding "I don't see Red Sea still at this moment as a negative."
- **The real exporter-side effect was time pressure.** Buyers were "hustling up to send the good sooner simply because the shipping right around Africa is taking longer", squeezing production windows. The Q3 FY24 investor presentation adds that alternative routes "resulted in delayed delivery, capacity crunch, and increased costs" across the industry.
- **The company names the Red Sea directly** in its FY24 annual report: "The Red Sea issue caused delays in shipments as cargo shipments had to resort to longer alternative routes."
- **Inventory days rose from 48 to 77** between FY23 and FY26, but growth, new capacity and US tariff uncertainty overlap, so the link to the Red Sea is not proven.
- **Exposure is rising.** Europe's share of sales more than doubled (3.3% to 7.8%) as the company diversified away from the US, putting more of its business on the Suez lane.

What management said, quarter by quarter

Table 5.5: Gokaldas Exports: management commentary by quarter

Quarter (months)	What management said	Link to the Red Sea
Q1–Q2 FY24 (Apr–Sep 2023)	No mention of shipping disruption	Before the crisis
Q3 FY24 (Oct–Dec 2023)	Red Sea "not a negative"; buyers want goods sooner; logistics cost borne by customers; some buyers airfreight	Direct, but cost passed on
Q4 FY24 (Jan–Mar 2024)	Airfreight at Atraco after production lost during the acquisition handover	None: production
Q1 FY25 (Apr–Jun 2024)	Election-period staff shortage: delays, cancellations, airfreight; ₹12.6 crore EBITDA hit; airfreight ₹3.4 crore (GEX) and ₹8.6 crore (Atraco)	None: production

Quarter (months)	What management said	Link to the Red Sea
Q2 FY25 (Jul–Sep 2024)	Spillover airfreight in India; Atraco airfreight until August 2024	None: production
Q3–Q4 FY25 (Oct 2024–Mar 2025)	Airfreight falls away	—
Q1–Q3 FY26 (Apr–Dec 2025)	Calls dominated by US tariffs (81–160 mentions of "tariff" per call)	—
Q4 FY26 (Jan–Mar 2026)	"The U.S./Iran war has impacted textile value chain with increased cost of raw materials like fuel, packaging, polyester and trims"; "Shipping costs have increased as well"	New Middle East shock

Source: Gokaldas Exports earnings-call transcripts and investor presentations, Q1 FY24 to Q4 FY26 (12 quarters, 36 documents).

A lesson in method. Gokaldas's airfreight peak (Q2 FY25, July–September 2024) fell in the same quarter as the freight-rate peak. Annual data alone would suggest the Red Sea caused it; the quarterly transcripts show it did not. Triangulating sources prevented a wrong conclusion.

Second company: KPR Mill Ltd

KPR Mill (Coimbatore), a vertically integrated yarn, fabric and garment maker, provides a second case. Its freight intensity rose modestly in the first crisis year and then eased, with no spike in FY25.

Table 5.6: KPR Mill: consolidated financial indicators, FY22 to FY26

Year	Consolidated sales (₹ cr)	Export share	Freight and forwarding (₹ cr)	Freight, % of sales	Freight, % of export sales	Change in inventories (₹ cr)
FY22	4,674.8	36.7%	33.5	0.72%	1.95%	+375.5 (build-up)
FY23	5,960.2	38.0%	46.6	0.78%	2.06%	+609.7 (build-up)
FY24	5,823.9	38.7%	53.1	0.91%	2.36%	+6.8
FY25	6,135.7	41.8%	48.9	0.80%	1.91%	−36.7 (release)
FY26	6,377.6	43.8%	55.2	0.87%	1.98%	−24.5 (release)

Sales = domestic + export sales; inventory change from the consolidated cash-flow statement. Source: KPR Mill annual reports FY2022-23 to FY2025-26 (₹ lakhs converted to ₹ crore).

- Freight intensity rose from 0.78% to 0.91% of sales in FY24, the first crisis year, then fell back to 0.80% in FY25 even though spot rates averaged 3.4 times pre-crisis.
- KPR's FY24 report says that adverse economic conditions, "coupled with the Red Sea crisis are impacting the export of Indian textile products". Its FY26 report lists "Red

Sea/Indian Ocean rerouting, higher shipping rates, and potential shipping disruptions" as supply-chain risks and states: "KPR mitigates via diversified sourcing, inventory buffers, and hedging."

- KPR built ₹610 crore of inventory in FY23, before the crisis, and released stock in FY25 and FY26, so there was no crisis-driven build-up.

Table 5.7: Freight as a percentage of revenue: Gokaldas Exports and KPR Mill

Freight as % of revenue	FY23	FY24	FY25	FY26
Gokaldas Exports (standalone)	0.92%	0.86%	1.62%	1.19%
KPR Mill (consolidated)	0.78%	0.91%	0.80%	0.87%

What the two companies show. For Indian exporters selling on FOB terms, the Red Sea crisis was mainly a buyer-side cost: the extra ocean freight was paid by European and US buyers. Supplier-side effects were time pressure and industry-wide demand effects, not a freight bill. This supports modelling the routing decision from the buyer's side (EuroThread, Section 7), and adds a supplier-side lesson: production disruptions such as labour shortages or acquisition integration can trigger more emergency airfreight than the route disruption itself.

5.5 Impact on inventory and working capital

Longer lead times raise inventory in two ways. Pipeline (in-transit) stock grows with transit days: 12 extra days on $60M of annual flow ties up about $2M more working capital. Safety stock grows with lead-time variability, which roughly triples on the Cape route (σ from 2 to 6 days), raising required safety stock almost threefold. Gokaldas's inventory days also rose (48 to 77), but growth, new capacity and tariffs overlap, and KPR Mill shows no comparable build-up, so that link is not proven.

5.6 Environmental impact

The Cape route is about 30–40% longer, and carriers sped up to protect schedules, so fuel burn and CO2 per container rose by an estimated 30–35% on affected voyages (UNCTAD). Under the EU Emissions Trading System, which covers shipping from 2024, this adds a carbon cost passed to shippers as surcharges.

5.7 Stakeholder impact

Table 5.8: Stakeholder impact

Stakeholder	Impact	Direction
Container carriers	Higher rates and capacity absorption boosted 2024 profits	Gain (short term)

Stakeholder	Impact	Direction
Importers and retailers (Europe)	Higher landed cost, stockouts, earlier ordering	Loss
Exporters (India, South and Southeast Asia)	Higher freight, cancelled or delayed orders, cash-flow strain for MSMEs	Loss
Egypt	Loss of a key source of foreign exchange	Major loss
Red Sea ports (Jeddah, Djibouti, Aqaba, Eilat)	Volumes fell sharply	Loss
Alternative hubs (Dubai, Tangier Med, Mediterranean, South Africa)	More sea-air and transshipment; congestion	Mixed
Insurers	Higher premiums, higher exposure	Mixed
Consumers	Modest price pressure on imported goods	Small loss
Seafarers	Direct safety risk; crew deaths in 2024	Severe loss
Environment	Higher emissions per voyage	Loss

5.8 Statistical tests

The shifts in traffic and freight rates are statistically significant, while freight rates show no negative effect on India's EU apparel export volumes.

Table 5.9: Before-and-after tests of transits and freight rates

Test (Welch t-test)	Before: 1 Jan – 18 Nov 2023	After: 1 Jan 2024 – 11 Oct 2026	Result
Bab el-Mandeb container transits per day	19.3 (322 days)	5.9 (1,015 days)	$t = -54.5$, $p < 0.001$, Cohen's $d = -3.9$
Suez Canal container transits per day	19.6	9.0	$t = -44.0$, $p < 0.001$, $d = -3.1$
Cape of Good Hope container transits per day	5.9	19.6	$t = 51.6$, $p < 0.001$, $d = 2.6$
Drewry spot rate, $ per FEU	1,468 (46 weeks)	3,625 (144 weeks)	$t = 15.9$, $p < 0.001$, $d = 1.9$

Mann–Whitney U tests, which do not assume normal distributions, give the same result (all $p < 0.001$). Monthly freight rates and Cape container transits are positively correlated ($r = 0.55$, $p < 0.001$). Effect sizes above 0.8 count as large, so these are very large changes.

Regression of India's monthly apparel exports to the EU (log, March 2023 – July 2026, $n = 41$) on the freight rate (log), with month-of-year dummies for seasonality:

Table 5.10: Regression of EU apparel exports on freight rates

Specification	β (freight elasticity)	Std. error (HAC)	p-value	R^2
Previous month's rate, with trend (main)	**+0.082**	0.019	< 0.001	0.93

Specification	β (freight elasticity)	Std. error (HAC)	p-value	R^2
Previous month's rate, no trend	+0.100	0.024	< 0.001	0.92
Same month's rate	+0.063	0.019	0.001	0.92
Rate two months earlier	+0.080	0.019	< 0.001	0.93
Year-on-year growth vs rate growth (n = 29)	+0.054	—	0.022	0.18

Interpretation. There is no evidence that higher freight reduced export volumes. The positive coefficient should not be read as freight causing exports: more likely, buyers ordered earlier and in larger lots when lead times stretched (front-loading), which raised both shipments and rates, a bullwhip-type effect (Lee et al., 1997). This supports the report's central point that the crisis is a cost and working-capital problem rather than a volume problem. With 41 observations and no control for European demand, the result is indicative, not causal.

CHAPTER 6
ROOT CAUSE ANALYSIS

The trigger is geopolitical, but the reason it became a large business problem is structural: global trade concentrated on one chokepoint, lean inventory policies with no buffer for lead-time shocks, and single-route logistics networks with no pre-planned alternatives.

6.1 Ishikawa (fishbone) diagram

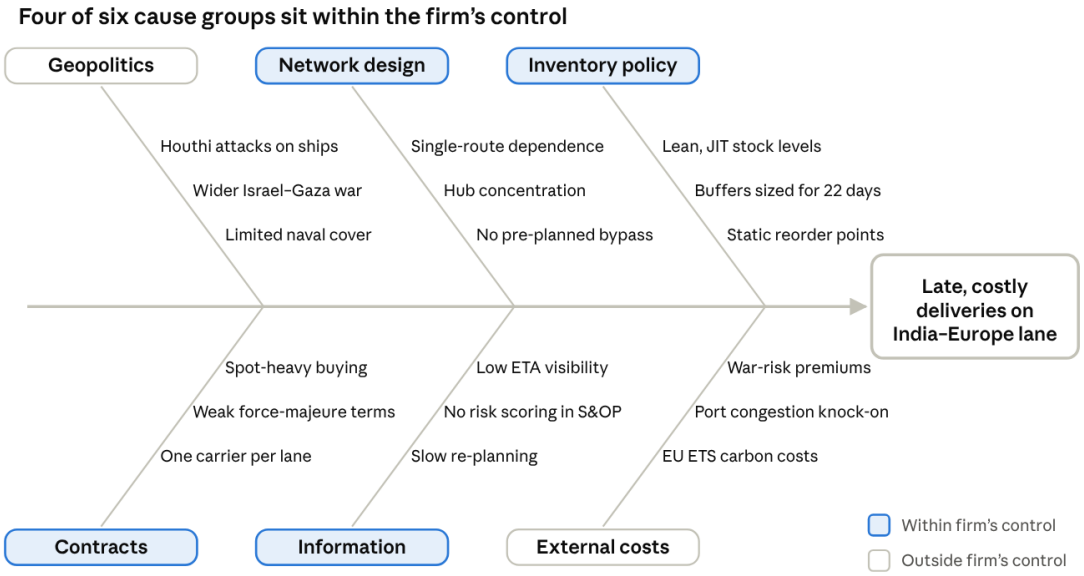


Figure 6.1: Ishikawa (fishbone) diagram of root causes
Source: Author's analysis.

The firm cannot stop the attacks, but network design, inventory policy, contracts and information are all within its control, which is where the solution must act.

6.2 Five Whys (from the case company's perspective)

1. **Why did EuroThread miss its spring season deliveries?** Containers arrived 2–3 weeks later than planned.
2. **Why were they late?** Carriers rerouted via the Cape, adding ~12 days, and schedules became unreliable.
3. **Why did a route change cause stockouts?** Order lead times and safety stock were set for a 22-day Suez transit with low variability.
4. **Why were parameters not adjusted quickly?** No routing-risk scenario was built into S&OP, and freight was bought on the spot market with no contingency clauses.

5. **Why was there no contingency?** Supply chain design optimised for cost under normal conditions; chokepoint risk was never quantified in the risk register.

Root cause: the absence of a risk-adjusted network and inventory design, not the attacks themselves, is what the firm can control.

6.3 PESTLE analysis

Table 6.1: PESTLE analysis

Factor	Drivers	Effect on supply chain
Political	Israel–Gaza conflict, Houthi strategy, US/EU naval responses, Iran's regional role	Persistent route insecurity; return depends on political settlement
Economic	Freight-rate spikes, war-risk premiums, Egypt's FX crisis, carrier overcapacity from 2025	Higher and more volatile landed cost
Social	Seafarer safety and union resistance to Red Sea transits	Carriers cannot return even if commercially attractive
Technological	Real-time AIS tracking, digital freight platforms, control towers	Enables faster rerouting and visibility
Legal	Force majeure clauses, war-risk insurance exclusions, sanctions compliance	Contract disputes; need for explicit disruption clauses
Environmental	Longer routes, higher speeds, EU ETS on shipping, IMO decarbonisation targets	Higher emissions and carbon costs

6.4 SWOT analysis of the industry response

Table 6.2: SWOT analysis of the industry response

	Helpful	Harmful
Internal	**Strengths:** large carrier fleets absorbed rerouting; digital visibility tools; growing fleet capacity from 2023–25 newbuild orders	**Weaknesses:** lean inventories; spot-heavy freight buying; single-route network design; weak risk quantification
External	**Opportunities:** multimodal corridors (sea-air via Gulf, IMEC, Middle Corridor); nearshoring to Turkey/North Africa; flexible index-linked contracts	**Threats:** escalation to other chokepoints (Hormuz); renewed attacks; congestion at alternative hubs; rising carbon costs

CHAPTER 7
COST MODEL: EUROTHREAD, A MODEL EUROPEAN BUYER (INDIA → HAMBURG)

At the latest 52-week average spot rate ($3,012 per FEU), the model buyer's landed logistics cost on the status-quo strategy is $7.52M a year, 2.5 times the pre-crisis $3.03M, and the largest increase is late-delivery losses, not freight.

7.1 Strategies modelled

Table 7.1: Strategies modelled

Code	Strategy	Description
S0	Pre-crisis baseline	Via Suez at $1,474 per FEU (Drewry, Jan–Nov 2023 average)
S1	Status quo: Cape spot	Via the Cape at $3,012 per FEU (Drewry, 52 weeks to 15 Oct 2026)
S2	Return to Suez with war-risk cover	Carrier-dependent; ~$2,360 per FEU + 0.5% cargo war-risk premium
S3	All sea-air via Dubai	Sea to Jebel Ali, air to Europe; ~$18,000 per FEU-equivalent, 12 days
S4	Cape on contract + EU buffer warehouse	Index-linked contract at ~$2,370 per FEU; bonded Hamburg 3PL holding 3–4 weeks of B/C cover
S5	Fixed hybrid	A-class always by sea-air; B/C via S4
S6	**Adaptive Resilience Strategy**	S4 + rule-based expediting + pre-agreed trigger to switch back to Suez
S7	Partial nearshoring	30% of B/C volume sourced from Turkey by road (+10% unit cost); rest via S4

7.2 Total landed logistics cost results (US$ million per year, 1,200 FEU)

Table 7.2: Total landed logistics cost by strategy (US$ million a year)

Strategy	Freight + insurance	Pipeline inventory	Safety stock	Late-delivery loss	Fixed (warehouse, premium)	Total	Per FEU ($)	CO2 (t)
S0 Pre-crisis Suez	1.77	0.72	0.13	0.41	0.00	**3.03**	2,520	1,920
S1 Cape spot (status quo)	3.61	1.12	0.34	2.45	0.00	**7.52**	6,270	2,592
S2 Return to Suez + war risk	3.13	0.72	0.23	0.98	0.00	**5.06**	4,220	1,920
S3 All sea-air	21.60	0.39	0.10	0.16	0.00	**22.25**	18,550	8,640
S4 Cape contract + EU buffer	2.84	1.12	0.34	0.65	0.18	**5.13**	4,270	2,592

Strategy	Freight + insurance	Pipeline inventory	Safety stock	Late-delivery loss	Fixed (warehouse, premium)	Total	Per FEU ($)	CO2 (t)
S5 Fixed hybrid	5.66	0.94	0.28	0.44	0.18	**7.50**	6,250	3,499
S6 Adaptive Resilience	2.84	1.12	0.34	0.65	0.23	**5.18**	4,320	2,592
S7 Partial nearshoring	3.03	0.92	0.30	0.57	1.55	**6.37**	5,310	2,151

7.3 Interpretation

- **Hidden cost dominates.** Against pre-crisis, S1's cost rises by $4.49M a year; freight explains only 41% of that ($1.84M) and late-delivery losses 45% ($2.04M). Firms that track only freight rates miss more than half of the crisis cost.
- **Contracting + buffering beats spot buying.** S4 saves $2.39M a year (32%) against S1, mainly by cutting the chance of a late delivery from 30% to about 8% and locking a lower rate.
- **Blanket expediting fails.** S3 and S5 buy speed the business does not need for most products.
- **Break-even expediting rule.** The sea-air premium is about $15,600 per FEU at current rates. For an A-class container ($80,000, 25% loss if late), expediting pays only if it cuts the chance of missing the window by more than 78 percentage points, that is, only for a container already known to be running late for a launch. For B/C goods it never pays.
- **Real-world check.** Gokaldas paid ₹20.3 crore of airfreight in FY25 (₹11.7 crore in Gokaldas Exports, ₹8.6 crore in Atraco) after labour and integration problems delayed production. Emergency airfreight is expensive whatever its cause; a rule-based approach limits it to shipments where it pays.
- **Option value.** S6 costs $0.05M a year more than S4 while the Cape route persists, but switches to Suez as soon as the trigger fires (Scenario B, Section 8.5).

7.4 Sensitivity analysis

Table 7.3: Sensitivity analysis of the annual saving

Case	Spot rate ($/FEU)	S1 total ($M)	S6 total ($M)	Annual saving ($M)
Base: 52-week average to 15 Oct 2026	3,012	7.52	5.18	**2.34**
Pre-crisis rate level	1,474	5.67	3.73	1.94
2025 average	2,627	7.06	4.82	2.24
2026 average to date	3,253	7.81	5.41	2.40

Case	Spot rate ($/FEU)	S1 total ($M)	S6 total ($M)	Annual saving ($M)
2024 average	4,941	9.83	7.00	2.83
Weekly peak, 18 Jul 2024	8,267	13.82	10.14	3.69
Base, late-delivery loss rates halved	3,012	6.29	4.85	1.44
Base, holding cost 30%	3,012	8.25	5.91	2.34
Base, warehouse cost doubled	3,012	7.52	5.36	2.16

S6 beats the status quo at every rate level Drewry recorded between January 2023 and October 2026, so the recommendation does not depend on where rates go next. The dashboard lets readers move the rate and loss assumptions themselves.

7.5 Monte Carlo simulation: how certain is the saving?

Across 10,000 simulated years with uncertain inputs, the Adaptive Resilience Strategy saves money in 99.99% of runs, with a median saving of $2.33M and a 90% range of $1.16M–$3.86M.

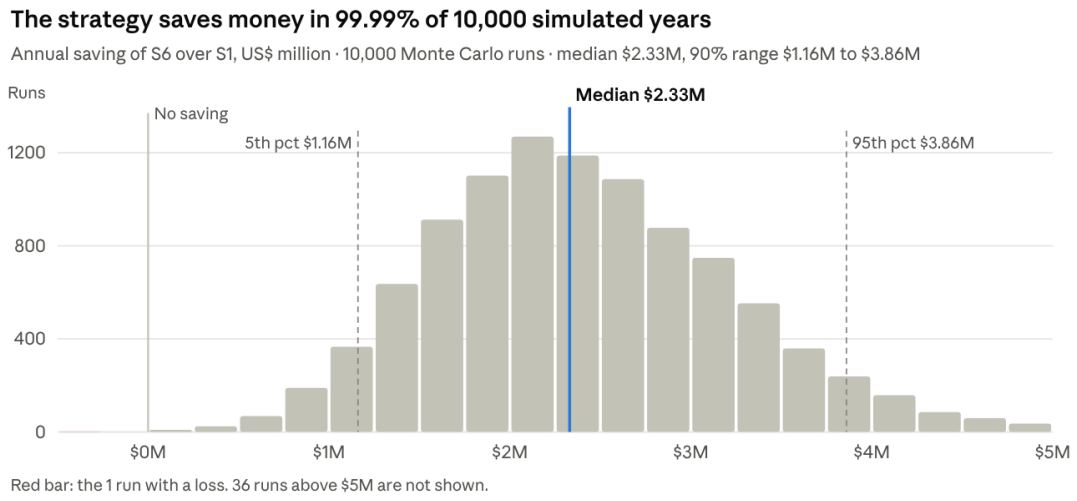


Figure 7.1: Monte Carlo distribution of the annual saving of S6 over S1 (10,000 runs)
Source: Author's simulation; spot rate bootstrapped from real Drewry data since January 2024.

Table 7.4: Monte Carlo input distributions

Input drawn at random each run	Distribution	Basis
Annual average spot rate	Bootstrap of 52 consecutive real Drewry weeks since January 2024 (5th–95th percentile $2,426–$4,887)	Real data
Chance of a late delivery, S1	Beta distribution, mean 30%	Assumption
Chance of a late delivery, S6	Beta distribution, mean 8%	Assumption
Late-delivery loss factor	Triangular, 0.5–1.0–1.5 × base	Assumption
Holding cost rate	Uniform, 15–25%	Assumption

Input drawn at random each run	Distribution	Basis
Cape lead-time standard deviation	Triangular, 4–6–9 days	Assumption
Contract rate	Uniform, 72–85% of spot	Assumption
Warehouse cost	Uniform, $150k–$250k a year	Assumption

Table 7.5: Monte Carlo results

Result ($M a year)	5th percentile	Median	95th percentile
S1 status quo, total cost	6.00	7.78	10.34
S6 Adaptive Resilience, total cost	4.28	5.30	7.24
Saving	**1.16**	**2.33**	**3.86**

The saving exceeds $1M in 97% of runs. Ranked by influence (Spearman correlation with the saving), the drivers are the chance of a late delivery under the status quo (0.70), the loss factor (0.45), the chance of a late delivery under S6 (−0.37) and the spot rate (0.28). Holding cost and lead-time variability barely matter. The most valuable data to collect next are therefore real late-delivery rates and markdown losses, for example from buyer chargeback and markdown records.

7.6 Optimal transport-mode mix (linear programming)

The optimiser confirms the recommendation and adds two practical rules: use sea-air only if the on-time target is above 91%, and negotiate for more contract capacity.

Table 7.6: Linear programming results: optimal transport-mode mix

Case	Feasible	Variable cost ($M)	Cape contract (FEU)	Cape spot (FEU)	Sea-air (FEU)	Suez (FEU)	CO2 (t)	Value of 1 extra contract FEU ($)
Cape only, OTIF ≥ 90%	Yes	4.81	960	240	0	0	2,592	823
Cape only, OTIF ≥ 91%	Yes	4.81	960	240	0	0	2,592	823
Cape only, OTIF ≥ 92%	Yes	6.02	960	144	96	0	3,076	5,881
Cape only, OTIF ≥ 93%	No	—	—	—	—	—	—	—
Suez reopened, OTIF ≥ 91%	Yes	3.41	0	240	0	960	2,054	2,139
Peak rate $8,267, OTIF ≥ 91%	Yes	10.04	960	240	0	0	2,592	1,948
CO2 cap 2,600 t, OTIF ≥ 91%	Yes	4.81	960	240	0	0	2,592	823

Variable cost covers freight, pipeline inventory and expected late-delivery loss; safety stock and fixed costs are the same in every case, so totals are lower than in Section 7.2. Constraints: contract capacity at most 80% of volume, sea-air at most 10%, late shipments at most (1 − OTIF target). The last column is the shadow price of the contract-capacity constraint.

- **Up to 91% OTIF, the cheapest plan uses no sea-air at all.** All A-class goods go on contract and 240 FEU of basics on spot.
- **Going from 91% to 92% OTIF costs $1.21M a year**, because 96 A-class FEU must then fly. The 92% target in Section 9.3 should be met by winning more contract capacity, not by flying goods; otherwise 91% is the efficient target. 93% is impossible within these capacity limits.
- **Contract capacity is worth real money:** each extra FEU on contract saves $823 at 91% OTIF and $5,881 at 92%.
- **The plan is robust:** at the 2024 peak rate the optimal mix is unchanged; only the cost rises.
- **When Suez reopens**, the optimiser moves all contract volume to Suez and cuts variable cost by 29% ($4.81M to $3.41M), which is the value the trigger playbook captures.
- **A CO2 cap of 2,600 t does not bind**: the optimal plan already emits 2,592 t.

7.7 Freight rate forecast

A statistical forecast puts the rate near $3,700 over the next six months, but its 80% range widens to $1,744–$8,018 by mid-January 2027, too wide to plan on. That is why the strategy relies on triggers rather than predictions.

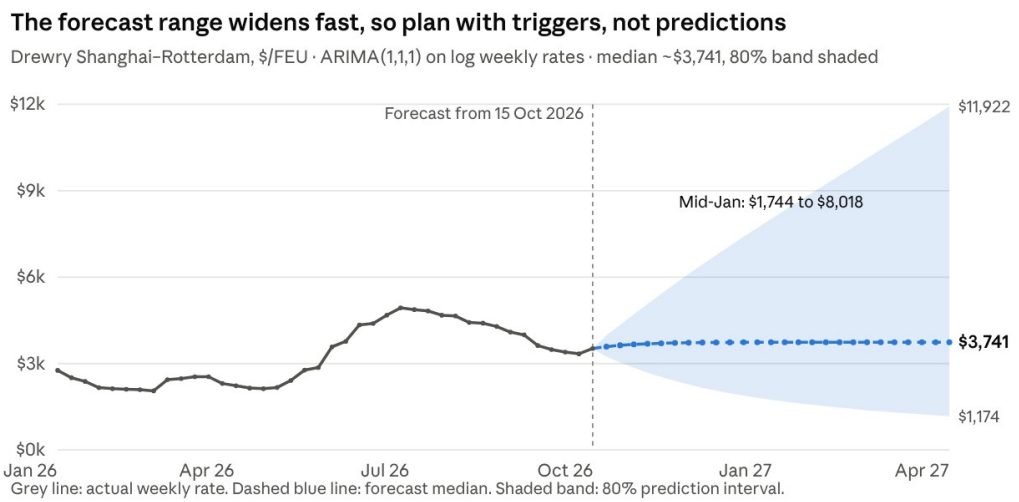


Figure 7.2: 26-week freight rate forecast with 80% prediction interval
Source: Author's ARIMA(1,1,1) model on log weekly Drewry rates, 5 January 2023 to 15 October 2026.

ARIMA models were fitted to the log of 195 weekly rates; ARIMA(1,1,1) had the lowest AIC (−350.9). Each model was then tested on the last 26 weeks, held out from fitting:

Table 7.7: Forecast back-test accuracy

Model	**Mean absolute % error over 26 weeks**
Naive (rate stays at its last value)	37.4%
ARIMA(0,1,1)	37.5%
ARIMA(1,1,0)	38.7%
ARIMA(1,1,1)	40.7%
ARIMA(2,1,1)	40.8%

No model beats the naive forecast. Freight rates move on shocks (attacks, ceasefires, regional conflict) that past values cannot predict, so a firm waiting for a reliable forecast will act too late. Monitoring leading signals (PortWatch transits, the weekly Drewry rate) and acting on pre-set triggers is the better approach.

CHAPTER 8
SOLUTION ALTERNATIVES AND EVALUATION

The Adaptive Resilience Strategy (S6) ranks first with a weighted score of 3.90 out of 5, ahead of partial nearshoring (3.55) and contract-plus-buffer alone (3.50).

8.1 Solution landscape

Responses exist at three levels; this project's recommendation operates at the firm level but draws on the other two.

Table 8.1: Solution landscape

Level	Solution	Status / feasibility
Firm (shipper)	Route diversification, contract strategy, buffer stock, product segmentation, nearshoring, digital visibility	Within the firm's control; evaluated below
Industry (carriers, ports)	Cape-based network redesign (e.g. Maersk–Hapag Gemini hub-and-spoke), sea-air via Gulf hubs, more capacity from newbuilds	Operating now
Policy and infrastructure	Naval escorts (Aspides, Prosperity Guardian); India–Middle East–Europe Corridor (IMEC); International North–South Transport Corridor (INSTC); Middle Corridor via Caspian; Northern Sea Route	Long-term or limited capacity; IMEC not yet operational

8.2 Evaluation criteria and weights

Weights reflect a fashion retailer's priorities and were set by pairwise comparison (AHP-style).

Table 8.2: Evaluation criteria and weights

Criterion	Weight	What a score of 5 means
Total landed cost	30%	Lowest annual TLC
Reliability (on-time availability)	20%	>95% of shipments in sales window
Lead time	15%	Shortest door-to-door time
Risk exposure and resilience	15%	Least exposure to security and single-route risk
Sustainability (CO2)	10%	Lowest emissions
Feasibility and control	10%	Implementable now, within firm's control

8.3 Weighted scoring matrix (scores 1–5)

Table 8.3: Weighted scoring matrix

Strategy	Cost (30%)	Reliability (20%)	Lead time (15%)	Risk (15%)	CO2 (10%)	Feasibility (10%)	Weighted score	Rank
S6 Adaptive Resilience	4	4	3	5	3	4	**3.90**	1
S7 Partial nearshoring	3	4	5	4	4	1	**3.55**	2
S4 Cape contract + EU DC	4	4	2	4	2	4	**3.50**	3
S2 Return to Suez + war risk	4	3	4	1	4	1	**3.05**	4
S5 Fixed hybrid	2	5	3	4	1	3	**3.05**	4
S3 All sea-air	1	5	5	4	1	2	**2.95**	6
S1 Cape spot (status quo)	2	2	2	4	2	5	**2.60**	7

8.4 Cost–benefit analysis of the recommended strategy (S6 vs status quo S1)

Table 8.4: Cost–benefit analysis of S6 against the status quo

Item	Year 1 ($)	Recurring per year ($)
One-time: warehouse onboarding, WMS/TMS integration, control-tower licence	250,000	—
Bonded 3PL warehouse (Hamburg)	180,000	180,000
Expedite allowance (rule-based sea-air)	50,000	50,000
Risk analyst (0.5 FTE) + training	60,000	60,000
Total cost	**540,000**	**290,000**
Gross saving vs S1 (freight, late-delivery loss)	2,570,000	2,570,000
Net benefit	**2,030,000**	**2,280,000**

Payback is about 2.5 months; the three-year NPV of net benefits at a 10% discount rate is about $5.4M, and year-1 ROI is about 375%. Savings use the real 52-week average rate of $3,012 per FEU.

8.5 Scenario analysis (2026–2027)

Table 8.5: Scenario analysis, 2026 to 2027

Scenario	Probability	Evidence	Spot rate assumed ($/FEU)	S1 TLC ($M)	S6 TLC ($M)
A. Prolonged diversion	40%	Cape still carries over 3× pre-crisis container traffic (PortWatch, 2026)	3,012	7.52	5.18

Scenario	Probability	Evidence	Spot rate assumed ($/FEU)	S1 TLC ($M)	S6 TLC ($M)
B. Gradual reopening	45%	Suez container transits at their highest since Dec 2023 (PortWatch, Oct 2026)	1,770	3.33	2.69
C. Escalation	15%	Mid-2026 rate spike to $4,933 (Drewry); wider regional conflict risk	6,000	11.10	8.00

The probability-weighted cost is about $4.5M a year for S6 against $6.2M for the status quo, a saving of about $1.7M. Probabilities are judgement-based; Scenario B was raised to 45% because of the September–October 2026 PortWatch signal. S6 is the best option in all three futures.

CHAPTER 9
RECOMMENDED SOLUTION: THE ADAPTIVE RESILIENCE STRATEGY

EuroThread should adopt S6, built on five pillars and implemented over 12 months. At current rates it saves about $2.3M a year while the Cape route persists, and it is set up to capture the Suez saving now that PortWatch shows traffic starting to return. For Indian exporters such as Gokaldas, the same logic applies on the supply side: spare production capacity so that labour or integration problems do not force emergency airfreight (as at Gokaldas in FY25), earlier production windows to meet buyers' earlier ship dates, and contract terms that share route-change costs with buyers.

9.1 The five pillars

1. **Contract portfolio.** Move 70–80% of volume to 12-month index-linked contracts (rate = index ± band) split across two carrier alliances; keep 20–30% on spot for flexibility. Add explicit route-change and force-majeure clauses.
2. **Strategic buffer at destination.** Use a bonded 3PL warehouse in Hamburg holding 3–4 weeks of B/C-class cover; replenish stores from it so a late ship no longer means an empty shelf. Recalculate safety stock monthly with live lead-time variance.
3. **Product segmentation (Fisher matrix + ABC/XYZ).** Basics (B/C, stable demand) go by contracted ocean freight; fashion and launch items (A, volatile) get earlier order cut-offs and are eligible for expediting.
4. **Rule-based expediting.** Sea-air via Dubai is used only when the break-even rule holds (avoided loss > freight premium), triggered automatically by the control tower when an A-class container's ETA slips past its launch date.
5. **Route-switch trigger playbook (real option).** Pre-agree with carriers and the 3PL the conditions for returning to Suez, so the switch happens within one booking cycle.

9.2 Suez return trigger playbook

Table 9.1: Suez return trigger playbook

Trigger level	Signal (data source)	Action
Green: switch	Bab el-Mandeb container transits above 12 a day for 8 weeks (PortWatch); no attacks for 90 days; two or more alliances back on Suez	Shift contracted volume to Suez routings; cut buffer to 2 weeks; renegotiate rate band
Amber: prepare	Suez container transits rising for 4+ weeks (PortWatch); carrier trials; war-risk premiums falling	Pilot 10% of B/C volume via Suez carriers; keep full buffer

Trigger level	Signal (data source)	Action
Red: hold or escalate	Any new attack on merchant shipping; spot rate up more than 30% in 4 weeks (Drewry)	Stay on Cape; raise buffer to 5 weeks; accelerate nearshoring study (S7)

Status in October 2026: Amber. Suez container transits have risen to 13.6 a day, but Bab el-Mandeb is still at about 5–6 a day, well below the Green threshold.

Implementation roadmap

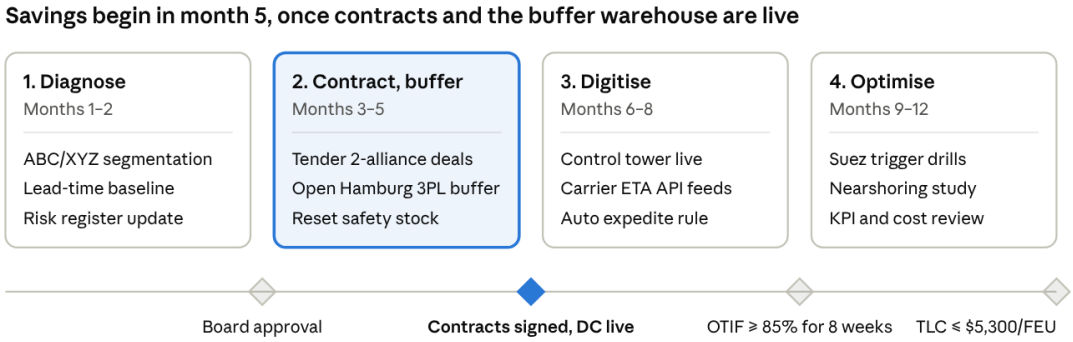


Figure 9.1: 12-month implementation roadmap
Source: Author's analysis.

Each phase starts only when the previous gate is met; phase 2 is where most of the saving is unlocked.

9.3 Key performance indicators

Table 9.2: Key performance indicators

KPI	Formula	Baseline (S1)	Target (12 months)
Total landed logistics cost per FEU	Annual TLC ÷ FEU shipped	$6,270	≤ $4,400
On-time in-full (OTIF) to stores	Orders on time and complete ÷ total	~70%	≥ 91% (≥ 92% only if contract share exceeds 80%; see 7.6)
Lead-time variability (σ, days)	Std. dev. of door-to-door days	6	≤ 3 (effective, with buffer)
Late-delivery loss, % of merchandise value	Markdowns + lost margin ÷ value	4.1%	≤ 1.2%
Contracted share of volume	Contract FEU ÷ total FEU	~0%	70–80%
Expedite share	Sea-air FEU ÷ total FEU	ad hoc	≤ 3%
CO2 per FEU	Tonnes CO2 ÷ FEU	~2.2 t	≤ 2.2 t (~1.6 t on Suez)
Route-trigger monitor	4-week average container transits, Suez and Bab el-Mandeb (PortWatch)	Suez 9.0, Bab el-Mandeb 5.5 a day (2026)	Green when Bab el-Mandeb > 12 a day for 8 weeks

KPI	Formula	Baseline (S1)	Target (12 months)
Time to switch after Green trigger	Days from trigger to first Suez booking	n/a	≤ 14 days

9.4 Risk register

Table 9.3: Risk register

Risk	Likelihood	Impact	Mitigation	Owner
Escalation closes Gulf hubs (sea-air via Dubai unavailable)	Low–Medium	High	Alternative sea-air via Colombo or direct air; buffer to 5 weeks	Head of Logistics
Spot rates collapse, making contracts look expensive	Medium	Medium	Index-linked bands, 20–30% spot share	Procurement
Buffer stock becomes obsolete (fashion risk)	Medium	Medium	Hold only B/C basics in buffer; monthly review	Merchandising
Carrier fails to honour contract capacity	Low	High	Split across two alliances; penalty clauses	Procurement
Data quality limits control-tower triggers	Medium	Medium	Carrier API integration; weekly data audit	IT / Analytics
Premature return to Suez followed by attack	Low	High	Three-level trigger, insurance check before switch	Risk Committee
Supplier cash-flow stress in India (MSMEs)	Medium	Medium	Supply-chain finance, earlier payment terms	Finance

CHAPTER 10
LIMITATIONS, CONCLUSION AND FUTURE SCOPE

10.1 Limitations

- Market data are real but have gaps: 14 of 195 Drewry weeks are carried forward or back-calculated, and Shanghai–Rotterdam is used as a proxy for India–Europe rates because no free India-specific weekly index was available.
- PortWatch estimates transits from AIS ship signals; it does not show cargo origin or voyage times.
- The model buyer's parameters (volume, product values, late-delivery loss rates) are assumptions. The ranking holds in every sensitivity case, but the annual saving ranges from $1.4M to $3.7M.
- Company evidence comes from two exporters (Gokaldas Exports and KPR Mill), both selling largely on FOB terms; the buyer-side cost model therefore relies on market data and assumptions rather than buyer accounts.
- Apart from the spot rate, the Monte Carlo input distributions are assumptions; the regression has 41 monthly observations and no control for European demand; and the forecast back-test shows rates cannot be predicted reliably from past values.

10.2 Conclusion

RQ1: Real data show container transits through Bab el-Mandeb down 72%, Cape transits up 234%, and Shanghai–Rotterdam rates at 3.4 times pre-crisis in 2024 and still 2.4 times in October 2026. **RQ2:** The shock spread because trade depends on one chokepoint and most firms ran lean, single-route, spot-bought networks; company evidence shows that Indian FOB exporters passed most ocean-freight cost to buyers but faced pressure to ship sooner, while production disruptions, not the route, drove Gokaldas's airfreight bill. **RQ3:** For a model European buyer, the Adaptive Resilience Strategy cuts total landed cost by 31% ($2.34M a year), pays back in about 2.5 months, and wins at every rate level observed since 2023. **RQ4:** With PortWatch showing Suez traffic starting to recover in September–October 2026, the trigger playbook lets firms return safely within one booking cycle, and raise buffers if the crisis escalates.

The wider lesson: resilience is cheaper to design in than to buy in a crisis. Firms that priced chokepoint risk into contracts, inventory policy and network design before 2023 paid far less for this disruption.

10.3 Future scope

- Collect real late-delivery and markdown data from company records to recalibrate the late-delivery loss rates, the model's most influential assumption.
- Extension to other chokepoints (Hormuz, Panama drought, Malacca) using the same framework.
- Feasibility study of IMEC and INSTC for India–Europe trade once operational.
- Primary research: interviews or surveys with Indian exporters (FIEO, Tiruppur Exporters' Association) to calibrate late-delivery losses.

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APPENDIX A
INTERACTIVE DASHBOARD

An interactive dashboard accompanies this report and presents the real data and analysis behind it. It is provided as a single HTML file:

Red_Sea_Strategy_Dashboard.html

The file opens in any web browser (Chrome, Edge, Safari or Firefox) and works without an internet connection. Keep it in the same folder as this report so the link opens it directly.

The dashboard has four tabs:

- Evidence: headline indicators, weekly freight rates, chokepoint transits, India’s apparel exports and the company comparison (Gokaldas Exports and KPR Mill).
- Strategies: cost breakdown, weighted scoring and comparison of all eight strategies.
- What-if: interactive selection of freight rate and late-delivery loss, with the resulting saving and the three scenarios.
- Advanced analysis: Monte Carlo distribution, freight forecast and linear programming results.

APPENDIX B
DATA AND DOCUMENTS USED

Table B.1: Primary data files and company documents

Source	Content	Coverage
IMF PortWatch	Daily chokepoint transit calls: Suez Canal, Bab el-Mandeb Strait, Cape of Good Hope	1 Jan 2023 – 11 Oct 2026
Drewry WCI	Weekly Shanghai–Rotterdam spot rate (195 weeks)	5 Jan 2023 – 15 Oct 2026
Ministry of Commerce, TradeStat	Monthly exports by country, HS 61, 62, 84, 85	Apr 2022 – Jul 2026
Gokaldas Exports Ltd	Annual reports (4); investor presentations, results and earnings-call transcripts (36)	FY2023 – FY2026
KPR Mill Ltd	Annual reports (4)	FY2023 – FY2026